

ABM BUSINESS LOGIC BIBLE

§8 — QUALITY CONTROL ENGINE (Module 6)

V2 Section 8 · The Gatekeeper · 7 QC Checks + Regeneration + Human Review SLA

§7 produces a message and a reasoning chain. §8 decides whether it may exist outside the engine. No message leaves the system without passing QC. V2 specifies a two-layer process: an AI QC Agent (Agent 7 from §7) validates first, then a human reviews the AI-approved message before it actually sends. §8 also now carries the validation surface that §7 v2 created — narrative-drift flags, Political Sensitivity gating, and Stakeholder Influence Strategy consistency — none of which existed when V2 was originally written.

QC checks	7 (V2) + 3 (§7 v2 bridge)	5-Whys chains	5 (root-caused)
New rules	29	Open questions	2 flagged

WHY §8 IS THE LAST GATE BEFORE THE IRREVERSIBLE ACTION

§9 (Outreach Execution) sends. §8 is everything standing between a generated message and a real Saudi bank executive's inbox. Every architectural discipline built across Sections 1–7 — grounded facts, RCM stamps, the Residual Uncertainty Register, vendor exclusivity, political sensitivity, narrative coherence — is only as good as §8's enforcement of it. A spec with perfect upstream rules and a weak QC gate still ships bad outreach.

8.0 THE TWO-LAYER GATE

AI QC (Agent 7, temperature 0.1) runs first and is cheap to iterate — failures regenerate automatically up to 3 times. Human review runs second and is mandatory during MVP for every AI-approved message, with no exceptions. The two layers catch different failure classes: AI QC catches mechanical violations (length, banned phrases, missing CTA); human review catches judgment calls AI cannot make.

Check ID	Check name	Pass criteria	Failure action
QC-001	Hallucination	Every factual claim traces to enrichment/signal data. No invented facts.	REJECT. Log claim. Re-queue Agent 6 with do-not-mention constraint.
QC-002	Tone	Enterprise B2B standard. No casual slang, no aggressive sales language, no desperation.	REJECT. Log phrase. Re-queue with tone guidance.
QC-003	Length	Email body ≤150 words. LinkedIn ≤80 words. Subject ≤60 chars.	REJECT. Log actual vs max. Re-queue.
QC-004	CTA	Exactly ONE clear, low-friction call-to-action.	REJECT if 0 or 2+. Log issue. Re-queue.
QC-005	Personalization	≥1 specific account/contact detail referenced.	REJECT if generic. Re-queue with context to include.
QC-006	Banned phrase	No banned phrases present (V2 §7.2 Agent 6 list).	REJECT. Auto-remove + re-validate.
QC-007	Compliance	No false Decimal claims, no outcome guarantees, no confidential info.	REJECT. ESCALATE to human — no auto-regenerate.

5-WHYS — WHY THIS QC ARCHITECTURE

Why #1: Why two layers (AI QC then human) rather than one?

Because the two layers catch genuinely different failure classes at genuinely different costs. AI QC is cheap, fast, and mechanical — it can check length and banned phrases in milliseconds and regenerate automatically. Human judgment is expensive and slow but catches what AI cannot: cultural tone-deafness, whether a politically-sensitive claim actually feels safe, whether the message reads as authentic in the specific KSA banking relationship context.

Why #2: Why does QC-007 (compliance) escalate to human rather than auto-regenerating like the other six checks?

Because compliance violations are not mechanical errors an AI regeneration can reliably fix. A model that hallucinated a false claim once may hallucinate a different false claim on the next attempt; the risk profile is asymmetric and legal/reputational, not just a quality miss. Human escalation is the only response proportionate to the risk.

Why #3: Why cap regeneration at 3 attempts rather than retry until it passes?

Because a message that fails QC three times on the same underlying issue indicates either an unsolvable input or a systemic prompt problem — not something a fourth automated attempt will fix. Capping at 3 routes the harder cases to a human who can diagnose root cause, consistent with §6 v3's stopping-condition discipline (ENR-STOP-02) applied to message generation.

Why #4: Why is human review mandatory for every AI-approved message during MVP, relaxed only on hard thresholds?

Because the cost of a bad outreach to a Saudi bank executive is asymmetric and reputational (§7 v2 AI-REP layer) — a single bad message can compound into market memory via SIG-MOBILITY. A threshold-based relaxation ($\geq 85\%$ approval, $< 10\%$ override) ships on demonstrated calibration, not on a calendar.

Why #5: Why must §8 validate against §7 v2's narrative, political, and stakeholder-strategy rules?

Because if §8 only enforces V2's original checklist, every rule added in §7 v2 becomes specified but never actually gated in production. §8 must grow with the architecture it protects.

Root cause: QC must be two-layered, risk-proportionate in escalation, bounded in retry cost, threshold-gated in relaxation, and continuously extended to enforce every constraint the architecture accumulates.

8.1 THE SEVEN V2 QC CHECKS — OPERATIONAL DETAIL

Rule ID	Specification
QC-RULE-001	Hallucination check: every claim must trace to a fact in Agent 2's brief, which traces to §6 dossier facts. Walks the citation chain, not just the message text.
QC-RULE-002	Additionally rejects any claim referencing a Residual Uncertainty Register item (§6 v3 ENR-BUDGET-03), even phrased as a question, except the acknowledge-as-question exception in AI-AG6-09.
QC-RULE-003	Tone check calibrated to KSA banking context: V2 §17 cultural rules bind (formal titles, no casual familiarity unless WARM/ACTIVE).
QC-RULE-004	Length check enforces 150/80/60 caps AND §7 v2 AI-ATTN-02 Attention-to-Value Ratio.
QC-RULE-005	CTA check validates friction-level against §7 v2 AI-TRUST-02: an aggressive CTA on low-trust contacts fails even with exactly one CTA present.
QC-RULE-006	Personalization check validates against §7 v2 AI-NARR-02: detail must be consistent with the Active Narrative.
QC-RULE-007	Banned-phrase check runs the V2 list plus any phrase added via §13 Loop 2 prompt-optimization feedback.

8.2 BRIDGE CHECKS — ENFORCING §7 v2's NEW CONSTRAINTS

Rule ID	Specification
QC-RULE-008	Narrative-drift check (new): message angle compared against Active Narrative (AI-NARR-01) and drift threshold (AI-NARR-03). Drift above threshold REJECTS and flags account for narrative-coherence review.
QC-RULE-009	Political Sensitivity gate (new): Medium/High rating (AI-POL-01) routes to mandatory human review, bypassing AI auto-approve AND any post-MVP relaxation. Hard gate, not regenerable.
QC-RULE-010	Stakeholder Influence Strategy consistency check (new): a Procurement Neutralization message that obscures a real constraint (violates AI-INF-02) FAILS and routes to QC-007 human escalation as a compliance-adjacent risk.

WHY THESE THREE ARE HARD GATES, NOT REGENERABLE FAILURES

QC-RULE-008/009/010 are properties of STRATEGY (Agent 5), not WORDING (Agent 6). Regenerating message text without revisiting the strategy brief cannot fix them. A failure here re-queues to Agent 5, a deeper re-queue than the standard QC failure path.

8.3 REGENERATION LIMITS

Rule ID	Specification
QC-REGEN-01	Maximum 3 regeneration attempts per message for Agent-6-level failures. After 3, flag MANUAL_CREATION — human writes from scratch using the AI's reasoning chain.
QC-REGEN-02	Each regeneration includes the specific QC failure reason as an explicit constraint.
QC-REGEN-03	Every regeneration logged: attempt_number, failure_reason, regenerated_output, new_qc_result.
QC-REGEN-04	Strategy-level failures (QC-RULE-008/009/010) re-queue to Agent 5 with a SEPARATE limit (max 2 attempts) before escalating to MANUAL_CREATION.

8.4 HUMAN REVIEW LAYER

Rule ID	Specification
QC-HUMAN-01	During MVP (first 90 days minimum): every AI-approved message goes to human review before sending. No exceptions.
QC-HUMAN-02	Human reviewer actions: APPROVE / EDIT / REJECT, logged with reason.
QC-HUMAN-03	SLA: reviewed within 4 business hours (9am–6pm KSA, Sun–Thu). Outside hours → next business day.
QC-HUMAN-04	Post-MVP relaxation: after 90 days, ≥85% AI approval AND <10% override → review optional for P3/P4. P1/P2 always reviewed.
QC-HUMAN-05	Relaxation evaluated per-segment AND per-language (resolves §7 OPEN-Q7.1): each cell crosses threshold independently.
QC-HUMAN-06	Medium/High Political Sensitivity (QC-RULE-009) NEVER bypasses human review regardless of relaxation status.
QC-HUMAN-07	Narrative-drift flags (QC-RULE-008) require human review of the flag itself — assessing whether the Active Narrative should change, not just whether to send this message.

8.5 OPEN-DESIGN-QUESTIONS — §8

OPEN-Q8.1 — STRATEGY-REGEN LIMIT INTERACTION WITH OPEN-Q7.2

Question: QC-REGEN-04 caps strategy-level regeneration at 2. §7's OPEN-Q7.2 already specifies an angle-conflict fallback chain. Do these interact — does each fallback step consume one of the 2 attempts, or are the budgets independent?

Candidate options: (a) Shared budget — simpler but may exhaust on angle resolution. (b) Independent budgets — OPEN-Q7.2's chain runs first with its own count. (c) Sequential — the whole OPEN-Q7.2 chain counts as ONE QC-REGEN-04 attempt.

Safe default: (c) Sequential. Conservative and easy to reason about.

OPEN-Q8.2 — WHO REVIEWS NARRATIVE-COHERENCE FLAGS

Question: QC-HUMAN-07's reviewer — the same routine QC reviewer, or the account_owner with full relationship context?

Candidate options: (a) Same QC reviewer — fastest, may lack context. (b) Routes to account_owner — right context, slower. (c) Decoupled — QC reviewer handles send/no-send; account_owner separately notified for the strategy question.

Safe default: (c) Decoupled, via the standard §11 handoff/notification pattern (INT-SLK).

§8 RULE ROLL-UP & CROSS-LINKS

Group	Rules	Key cross-links
7 V2 checks (detail)	QC-RULE-001..007	API-09, §6 v3 Residual Uncertainty Register, V2 §17, §7 v2 AI-ATTN/AI-TRUST
Bridge checks (new)	QC-RULE-008..010	§7 v2 AI-NARR-03, AI-POL-01/02, AI-INF-02 — hard gates
Regeneration	QC-REGEN-01..04	§6 v3 ENR-STOP-02, Agent 5 vs Agent 6 re-queue distinction
Human review	QC-HUMAN-01..07	§7 OPEN-Q7.1, AI-POL-02 permanent gate, INT-SLK notification
Open questions	2 flagged	A4 governance per TIER-OPEN-DISCIPLINE-02

§8 COMPLETE — WHAT IT ESTABLISHED

Quality Control is now the last gate before the engine's only irreversible action, fully specified at V2's operational level and extended to enforce every constraint accumulated through §7 v2. Three new bridge checks (narrative drift, political sensitivity, stakeholder-strategy consistency) are hard gates that re-queue to Agent 5, not Agent 6. Regeneration is bounded and risk-proportionate; human review is threshold-gated with Political Sensitivity and narrative-drift permanently exempted from relaxation. 29 new rules; 2 open questions.

§8 is the proof that every rule built in Sections 1–7 matters in production: a rule §8 doesn't enforce is a rule that doesn't exist when the message sends.

PLACEMENT MAP UPDATE

§8 status: **Complete-with-Opens** (OPEN-Q8.1, OPEN-Q8.2).

ON THE HORIZON

§9 Outreach Execution Engine — channel strategy, V2 email/LinkedIn sequence design, sequence execution rules, deliverability rules — bridged to INT-EML/INT-LIN and §7 v2's Portfolio Role + Trust-gated CTA discipline.